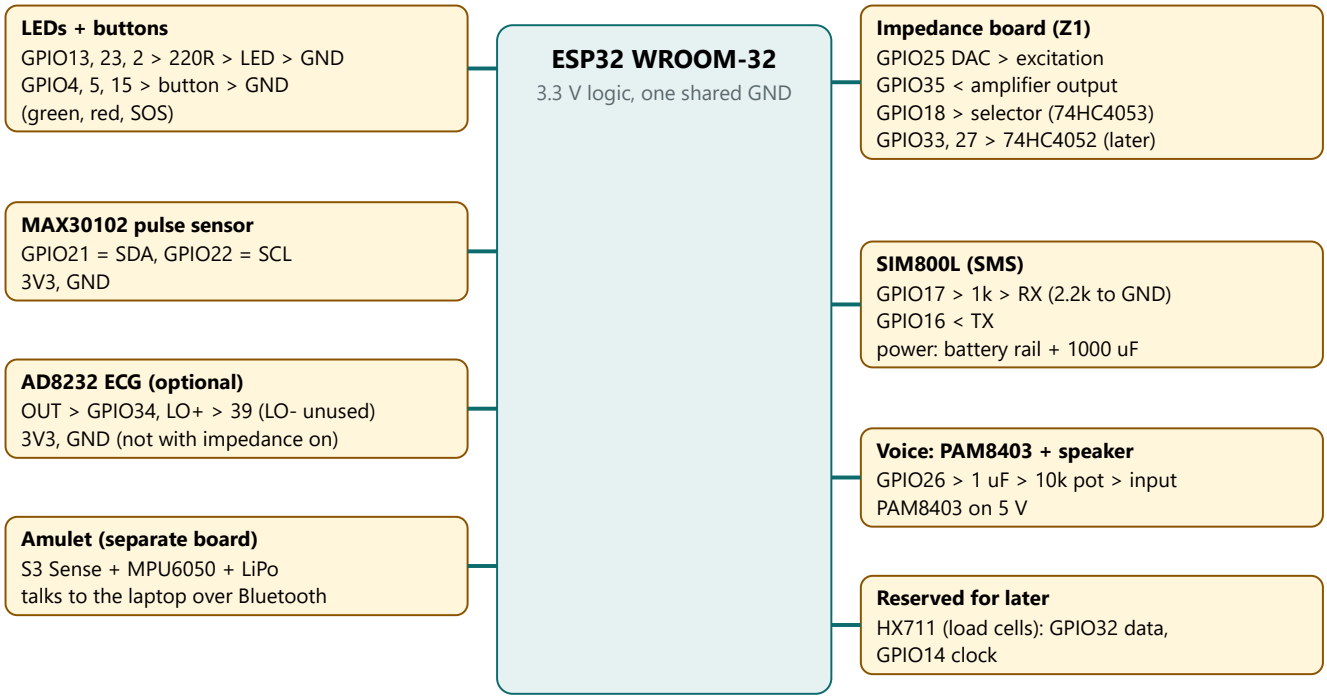
HridayMitra: wiring guide (no load cell yet)

Monitoring only, not a diagnosis tool. Pin numbers are the ESP32 DevKit GPIO numbers. Pin numbers on the ICs follow the standard layouts: **check each datasheet before you power up.**

Golden rules. 1) All logic is 3.3 V and all grounds join at one point. 2) Wire one block, run its test, then the next. 3) Put a 100 nF capacitor across the power pins of every IC, close to the chip. 4) Power off before you change wires. 5) Never connect the impedance board to a person until the resistor phantom test passes, and never to anyone with a pacemaker or defibrillator. 6) When a person touches the electrodes, run from battery or USB from a laptop on battery, not a mains charger. 7) Wire colours: red = 3.3 V, black = GND, yellow/white = signals, orange = 5 V, blue = battery.

1. Overview



2. Power

- **For now: USB.** Plug the ESP32 into the laptop. Its 3V3 pin powers the MAX30102, impedance board, AD8232 and LEDs. This is the safest way to start.
- **Battery version (later):** 18650 cell → TP4056 (B+ and B-) → the TP4056 OUT+ / OUT- is the **battery rail** (3.7-4.2 V). Boost module input from the battery rail, output set to 5 V → ESP32 VIN and PAM8403. SIM800L takes the battery rail directly (not 3V3, not 5 V) with a 1000 µF capacitor across its power pins.

Rail	Where it comes from	What it powers
3.3 V	ESP32 3V3 pin	MAX30102, AD8232, MCP6002/6022, INA126, 74HC4053/4052, LED and button logic
5 V	USB (VIN pin) or boost module	PAM8403 amplifier
Battery rail 3.7-4.2 V	TP4056 OUT	SIM800L only (needs up to 2 A bursts)
GND	All grounds joined at the ESP32 GND	Everything

3. Pin map

ESP32 pin	Goes to	Notes
GPIO13, 23, 2	LED green, amber, red	each through 220 Ω to GND
GPIO4, 5, 15	Buttons green, red, SOS	other leg of each button to GND; internal pull-ups
GPIO21 (SDA), GPIO22 (SCL)	MAX30102	3.3 V, GND
GPIO25 (DAC1)	Impedance excitation	5 kHz sine
GPIO35 (ADC1)	Impedance amplifier output	input only
GPIO18	74HC4053 select (A and B tied)	low = body, high = reference (across Rcal when GPIO12 is high)
GPIO33, GPIO27	74HC4052 path select S0, S1	only for the trunk paths, later
GPIO26 (DAC2)	Voice to PAM8403	through 1 µF and 10 kΩ pot
GPIO17 (TX2), GPIO16 (RX2)	SIM800L	TX through a divider (see section 8)
GPIO34, 39	AD8232 OUT, LO+	inputs only; optional
GPIO19	Seat vibration motor	1 k to 2N2222 base; motor 5 V to collector; 1N4148 across motor
GPIO36	Tablet-drawer microswitch	switch to GND; 10 k pull-up to 3V3 (no internal pull-up)
GPIO12	Self-calibration select (74HC4053 S3)	boot pin: only this high-impedance input, no pull-up
GPIO32, GPIO14	Reserved: HX711 data, clock	when the load cells arrive
GPIO0, GPIO12	**Leave empty**	boot pins

4. Block 1: LEDs and buttons (test first)

From	To
GPIO13 → 220 Ω → green LED long leg	LED short leg → GND
GPIO23 → 220 Ω → amber LED	short leg → GND
GPIO2 → 220 Ω → red LED	short leg → GND
GPIO4, GPIO5, GPIO15	each to one leg of a button; the other leg to GND

Test: a sketch that blinks each LED and prints the button on the Serial Monitor. Pass: each LED lights, each press prints.

5. Block 2: MAX30102 pulse sensor

Module pin	Connect to
VIN	3V3 (if your module says 3.3-5 V; some need 5 V: read the board)
GND	GND
SDA	GPIO21
SCL	GPIO22
INT	not used

Test: an I2C scanner shows address `0x57`; then a MAX30102 example prints a heart rate when you rest a finger on it. Pass: it matches your wrist pulse.

6. Block 3: the impedance board (legs path Z1)

Build on the breadboard from **left to right**: Vref and op-amp, selector, amplifier. Keep the DAC wire (GPIO25) away from the ADC wire (GPIO35). Parts: U1 = MCP6002 (or MCP6022), U2 = INA126, U3 = 74HC4053. Standard dual op-amp pins: 1 OUTA, 2 INA-, 3 INA+, 4 V-, 5 INB+, 6 INB-, 7 OUTB, 8 V+. INA126 DIP-8: 1 RG, 2 -IN, 3 +IN, 4 V-, 5 REF, 6 OUT, 7 V+, 8 RG (verify).

Net	Connect
Power	U1 pin 8, U2 pin 7, U3 VCC → 3V3. U1 pin 4, U2 pin 4, U3 GND, VEE, INH and select C → GND. 100 nF across each IC's power pins; 10 µF across the 3V3 rail.
VREF (1.65 V)	3V3 → R1 10 kΩ → node VREFIN → R2 10 kΩ → GND. C1 10 µF from VREFIN to GND. VREFIN → U1 pin 3. U1 pin 1 tied to U1 pin 2. U1 pin 1 is **VREF**.
Excitation	GPIO25 → C2 1 µF → node EXC_IN. R3 100 kΩ from EXC_IN to VREF. EXC_IN → U1 pin 5. U1 pin 7 tied to U1 pin 6. U1 pin 7 is **EXC_OUT**.
Current path	EXC_OUT → C3 1 µF → R4 4.7 kΩ → **FORE_L** (left forefoot plate). **FORE_R** (right forefoot plate) → C4 1 µF → node **RET** → R5 120 Ω → node **NCAL** → **Rcal 100 Ω 0.1%** → VREF. 74HC4053 switch C: C0 = RET, C1 = NCAL, common **RTAP** → switch A input A1; select S3 = GPIO12 (self-calibration).
Sensing	**HEEL_L** → R6 1 kΩ → node SA. **HEEL_R** → R7 1 kΩ → node SB. R8 10 MΩ from SA to VREF. R9 10 MΩ from SB to VREF (10 M, not 1 M: see Monte Carlo).
Selector U3 (74HC4053)	Select pins A and B tied together → GPIO18. X0 = SA, X1 = RET, X common → INA126 pin 3 (+IN). Y0 = SB, Y1 = VREF, Y common → INA126 pin 2 (-IN). (Use your datasheet's names for these pins.)
Amplifier U2 (INA126)	RG 16 kΩ (15 kΩ + 1 kΩ) between pins 1 and 8. REF pin 5 → VREF. OUT pin 6 → R11 1.5 kΩ → node ADC → **GPIO35**. C5 10 nF from ADC node to GND.

Test 6a: voltages with a multimeter (no plates connected)

- VREF reads about **1.65 V**. EXC_IN and EXC_OUT also about 1.65 V (DC).
- Short INA pin 2 and pin 3 together: OUT (pin 6) reads about 1.65 V.

Test 6b: the resistor phantom (instead of legs)

No plates yet? Use a resistor: put **330 Ω** between FORE_L and FORE_R. Wire HEEL_L to the FORE_L end and HEEL_R to the FORE_R end. Run the measurement: Z should read about 330 Ω. Repeat with 680 Ω and 1 kΩ (within about 5%). Then add 1-5 kΩ in series with each outer lead: Z should barely move.

7. Block 4 (optional): AD8232 ECG

Module pin	Connect to
3.3V	3V3
GND	GND
OUTPUT	GPIO34
LO+	GPIO39
SDN	leave open

Electrode leads: RA (red) right side, LA (yellow) left side, RL (green) reference. **Do not connect the ECG leads and the impedance board to the same electrodes at once.** Test: plot GPIO34 on the Serial Plotter; you should see heartbeats.

8. Block 5: SIM800L (SMS)

Module pin	Connect to
VCC	Battery rail (3.7-4.2 V) with 1000 µF across VCC-GND. Not 3V3.
GND	GND (common)
TX	GPIO16
RX	GPIO17 → 1 kΩ → RX, and RX → 2.2 kΩ → GND (a divider, about 2.3 V)
Antenna	Screw on the antenna; SIM card in (Airtel, Vi or BSNL)

Test: send `AT` at 9600 baud: it answers `OK`. `AT+CSQ` shows signal; then send one SMS to your own phone. If it resets while sending, the supply is too weak: add the capacitor, shorten wires, use a better 18650.

9. Block 6: voice (PAM8403 and speaker)

From	To
GPIO26	1 µF capacitor → 10 kΩ potentiometer top
Pot bottom	GND
Pot wiper	PAM8403 L input (R input to GND)
PAM8403 5 V, GND	5 V rail, GND
PAM8403 L output + and -	Speaker (4 Ω 3 W)

Test: play one short clip from flash. Pass: clear and loud with the pot about half. If it hums, shorten the input wire and keep it away from the SIM800L.

10. Block 7: amulet (separate board)

Example for a Seeed XIAO ESP32S3 Sense (check your board's pin names): MPU6050 VCC → 3V3, GND → GND, SDA → D4 (GPIO5), SCL → D5 (GPIO6), AD0 → GND (address 0x68). SOS button between D1 and GND. LiPo battery to the BAT+ and BAT- pads on the underside (**check polarity**). **Test:** an I2C scan finds 0x68; the tilt angle changes when you tilt the board against a protractor.

11. Order and quick checklist

#	Block	Pass when
1	ESP32 and USB	Serial Monitor prints
2	LEDs and buttons	Each LED and button works
3	MAX30102	0x57 found; pulse matches wrist
4	Impedance: voltages (6a)	VREF 1.65 V; INA shorted = 1.65 V
5	Impedance: phantom (6b)	330, 680, 1 kΩ within about 5%
6	SIM800L	AT = OK; SMS arrives
7	Voice	Clear and loud
8	Amulet	0x68 found; tilt works
9	ECG (optional)	Heartbeats visible

12. If something misbehaves

Symptom	Check
ESP32 won't boot or upload	Anything wired to GPIO0 or GPIO12; unplug the board from the breadboard to flash
I2C scan finds nothing	SDA and SCL swapped; module powered; pull-ups
Impedance stuck at one value	Selector pins A and B tied to GPIO18; INA REF at VREF; no short to GND
Impedance jumps around	Loose breadboard contact; long wires near the DAC; no 100 nF decoupling

Reads depend on contact	R8 and R9 10 MΩ missing or 1 M fitted by mistake; the 1 kΩ protection resistors missing
SIM800L resets	Weak supply; add the 1000 µF capacitor
Hum in the speaker	Shared ground wiring; shield the input; separate it from the SIM800L

Not verified: IC pin numbers (check the datasheets), the pin map on your exact board, and all values (starting values from the simulation).